2016-Summer Paper 3 – Subspeciality of Urology

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Q1. Discuss minimally invasive and endoscopic management of upper urinary tract malignancy.

Type: LAQ (25 marks)

Introduction

- **UTUC (upper tract urothelial carcinoma)** is 5–10% of urothelial cancers. Bladder recurrence follows in 22–47%, mostly from downstream seeding.
- Standard is **RNU (radical nephroureterectomy) with bladder cuff**; kidney-sparing versus radical surgery is chosen by **risk**.
- URS (ureteroscopy) biopsy is small and shallow, so it **cannot stage**. Stage is inferred from grade, endoscopic appearance, imaging and cytology together.

Risk stratification

EAU (European Association of Urology)

Low risk (all present)	High risk (any one)
Unifocal	Multifocal
Under 2 cm	2 cm or more
Low-grade cytology and biopsy	High-grade cytology or biopsy
No invasion on CTU (computed tomography urography)	Hydronephrosis; invasion on CT; prior radical cystectomy; variant histology

- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023** splits each into favourable and unfavourable (obstruction, multifocality, lower-tract involvement). Ablation is **preferred** in low-risk favourable, **offered** in low-risk unfavourable, **rare** in high-risk favourable, **palliative** in high-risk unfavourable. Endoscopic therapy suits tumours **under 1.5 cm**.

Indications for kidney-sparing treatment

- **Elective**: low-risk disease, whatever the other kidney.
- **Imperative**: solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit for RNU, after shared decision.
- The patient must accept frequent recurrence and **lifelong endoscopic surveillance**.

Ureteroscopic (retrograde) management

- Closed system, lowest morbidity; for small ureteric and renal tumours.
- **Steps**: cystoscopy → selective cytology and saline barbotage → no-touch semirigid then flexible URS → map all lesions → biopsy (3 Fr forceps or basket) into fixative → ablate the base with laser → stent optional.

Laser	Penetration	Note
Ho:YAG (holmium:yttrium-aluminium-garnet)	Under 0.5 mm	Safe in thin ureter
Nd:YAG (neodymium:YAG)	Up to 5–6 mm	Coagulative necrosis; transmural injury risk
Thulium	Under 0.2 mm	Strong haemostasis

- Avoid circumferential fulguration (stricture); stage large tumours.
- Results: upper-tract recurrence 23–90%, bladder recurrence 15–45%, progression 0–20%. **Grade is the strongest predictor.**

Percutaneous (antegrade) management

- For large pelvic tumours, lower-pole or inaccessible calyces and urinary diversions.
- **Steps**: retrograde catheter → puncture a calyx away from the tumour (upper or middle calyx for pelvic tumours) → dilate to 30 Fr → cold-cup, bipolar resection or laser; base sent separately → nephrostomy left → **second-look nephroscopy**.
- Removes more tumour and can **stage**; the tract allows topical therapy. But it is an **open system**: tract seeding about 0.3%, bleeding.

Segmental ureterectomy

- **Distal ureterectomy with bladder cuff and reimplantation** (psoas hitch or Boari flap) for distal tumours, including high-grade ones when the kidney must be saved. It gives a staging specimen; add pelvic lymphadenectomy if high grade.
- Mid-ureteric resection (frozen-section margins) is rare. Laparoscopic or robotic routes are feasible; outcomes equal RNU in selected patients.

Laparoscopic and robotic RNU

- **Principles**: clip the ureter early; never open the collecting system; remove the **whole intramural ureter and cuff**; bag extraction intact; watertight bladder closure; template lymphadenectomy for high risk.
- **Steps**: modified flank, 3–4 ports → reflect colon → clip ureter → artery then vein → kidney outside Gerota's fascia (adrenal spared) → ureter traced under the superior vesical artery → cuff excised (extravesical or open) → two-layer closure.
- Abandoned (higher recurrence): transurethral "pluck", stripping, laparoscopic staple.
- Oncologically **equivalent to open** (AUA/SUO), with less blood loss and shorter stay; less favoured in bulky T3/T4. Robotic yields more nodes.

Adjuvant topical therapy

- **Single postoperative intravesical dose** (AUA/SUO strong): **ODMIT-C (2011)** MMC (mitomycin C) 40 mg, 28% → 16% bladder recurrence at 1 year; **Ito (2013)** pirarubicin, 42% → 17% at 2 years.
- **Upper-tract instillation** via nephrostomy (low pressure, 25 cm H2O), ureteric catheter or reflux up a double-J. **BCG (bacillus Calmette–Guérin)**: 6-week course for upper-tract CIS

(carcinoma in situ), and after complete ablation of high-risk favourable Ta/T1. Risks: sepsis, stricture.

- **UGN-101 (Jelmyto)**: mitomycin reverse-thermal gel, **FDA (Food and Drug Administration) 2020** for low-grade UTUC not manageable endoscopically. **OLYMPUS**: complete response 59%, 56% disease-free at 12 months; ureteric stricture about 44%.

Systemic therapy

- **POUT (2020)**: adjuvant platinum after RNU for pT2–T4 or N+: 3-year disease-free survival 71% vs 46%. Neoadjuvant platinum if renal function is borderline (RNU removes cisplatin eligibility).

Surveillance (AUA/SUO)

- **Kidney-sparing**: cystoscopy and URS within 3 months, URS again at 6 and 12 months; imaging 3–6-monthly (high risk) or 6–9-monthly (low risk), then yearly to 5 years.
- **After RNU**: cystoscopy and cytology from 3 months; CT yearly (more often if pT2 or more).

Recent advances

- Lynch syndrome screening; thulium fibre laser; single-use digital ureteroscopes.

Diagram to draw: ureteroscopic laser ablation of a pelvic tumour beside RNU with bladder cuff.

> **Key points to remember:**
>
> - EAU low risk needs all: unifocal, under 2 cm, low grade, no invasion; any single adverse feature makes it high risk.
> - Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
> - Single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).

Q2. Laparoscopic pyeloplasty.

Type: LAQ (25 marks)

Introduction

- **PUJO (pelviureteric junction obstruction)** is impaired flow from pelvis to ureter that, if untreated, causes progressive renal damage (Koff).
- Causes: **intrinsic** adynamic segment (abnormal circular muscle), **extrinsic** crossing lower-pole vessel, high insertion; secondary (failed repair, stone).
- **Anderson–Hynes (1949) dismembered pyeloplasty** is the gold standard by any route. First laparoscopic pyeloplasty **1993 (Schuessler)**; robotic **2002**. Success **over 90–95%**, equal to open, with less morbidity.

Indications

- Symptoms (pain, Dietl's crisis), infection, stones, hypertension (rare).
- Impaired split function or **fall of 10% or more** on serial renography; rising APD (anteroposterior diameter); obstructed solitary kidney or bilateral disease.
- Children: most antenatal hydronephrosis resolves; about one-third need surgery.
- **Contraindications**: uncorrected coagulopathy, untreated UTI (urinary tract infection), cardiopulmonary disease barring pneumoperitoneum.
- **Nephrectomy instead** if split function under 15% in adults with a normal other kidney.

Preoperative evaluation

- **USG (ultrasonography)**: large pelvis, small calyces, ureter not seen.
- **MAG3 (mercaptoacetyltriglycine) diuretic renogram**: split function and an obstructive curve that does not wash out after furosemide. Usual teaching: washout half-time under 10 minutes normal, 10–20 minutes equivocal, over 20 minutes obstructed (Ch 93 quotes over 40 minutes). Split function under 40% supports repair.
- **CT (computed tomography) angiography**: anatomy, **crossing vessels**, stones.
- **Retrograde pyelogram** at surgery: confirms the level, excludes ureteric stricture.
- **Whitaker test** if equivocal: perfusion 10 mL/min; pelvic pressure over 15–20 cm H2O means obstruction.
- Sterile urine. Drain beforehand (stent or nephrostomy) only for infection, azotaemia or severe pain.

Approaches

	Transperitoneal	Retroperitoneal
Space	Large, familiar landmarks	Small, created by balloon
Bowel	Colon reflected (left: **transmesocolic** window)	Not entered; leak stays extraperitoneal
Use	Commonest; easier suturing	Avoids bowel; harder orientation

- One paediatric randomised trial favoured retroperitoneal (shorter stay); overall outcomes similar.

Steps (transperitoneal Anderson–Hynes)

1. Cystoscopy, retrograde pyelogram, catheter.
2. 45–60° flank; pneumoperitoneum 12–15 mmHg in adults (under 12 mmHg in children); 3–4 ports, camera at umbilicus.
3. Reflect colon medially, or open the mesocolon over the pelvis.

4. Find the ureter; trace it to the PUJ with its periureteric tissue; minimal diathermy. Look for a **crossing vessel**.
5. Hitch stitch on the pelvis; transect the pelvis above the PUJ; **reduce** a redundant pelvis if needed.
6. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a handle (no-touch).
7. **Transpose** the PUJ in front of any crossing vessel.
8. Anastomose the spatulation apex to the most dependent pelvis with 4-0 absorbable (5-0/6-0 in infants): back wall, stent, front wall, then close the pelvis. Watertight, tension-free, funnel-shaped.
9. **Antegrade double-J** over a wire; clamp the catheter: urine from the stent side holes confirms the bladder coil.
10. Excise the narrow segment; drain near (not on) the suture line.

Crossing vessel

- It lowers endopyelotomy success (42% vs 86%), so pyeloplasty is preferred. **Never divide** a lower-pole vessel (segmental infarction).
- **Vascular hitch (Hellström)**: vessel moved up onto the pelvis, only if the PUJ is intrinsically normal and peristalsing.

Stenting and postoperative care

- Double-J for **4–6 weeks** (children 1–4 weeks: no difference). Externalised stent in infants avoids a second anaesthetic.
- Foley for 24–48 hours (bladder pressure refluxes up the stent); drain out when dry. If drain output rises after Foley removal, replace the Foley.
- **MAG3 renogram** after stent removal, then ultrasound; follow for years.

Robotic pyeloplasty

- 3D vision, wristed suturing, tremor filtration, **shorter learning curve**. Now first-line at major centres.
- Success 94.7–100%, the same as laparoscopy; advantage is ergonomic, at higher cost.

Outcomes

- Laparoscopic: operating time 2–4 hours; complications 2–16%; conversion 0–5.5%; stay about 2.7 vs 4.2 days (open); success 87–99%.
- **Up to 30% of failures appear after 2 years**, so follow-up must be long.

Complications

- Urine leak and urinoma, stent symptoms or migration, UTI, haematuria, ileus, bowel or vessel injury, port-site hernia, recurrent obstruction.

Management of failure

```
Early (clot, oedema)        -> nephrostomy, wait
Persistent obstruction      -> CT angiography (missed vessel?)
  short, no vessel          -> endopyelotomy (about 70%)
  otherwise                 -> redo laparoscopic/robotic pyeloplasty
                               (about 89% after failed open; buccal graft onlay)
  small intrarenal pelvis   -> ureterocalicostomy
  poor function             -> nephrectomy
```

Also know: Y-V plasty (high insertion); Culp spiral flap (long narrowing); stones removed through the pyelotomy with a flexible scope; safe in horseshoe and pelvic kidneys.

Recent advances

- Single-port robot (2018); indocyanine green fluorescence for redo cases; buccal mucosa ureteroplasty.

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition over the crossing vessel, dependent anastomosis.

> **Key points to remember:**
>
> - Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
> - Transpose over a crossing vessel; never divide it.
> - Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
> - Robotic equals laparoscopic in outcome, with easier suturing.

Q3a. Oligoanaesthospermia.

Type: SAQ (10 marks)

Assumed meaning: oligoasthenozoospermia (low sperm count with poor motility).

Definition (WHO (World Health Organization) 2021, 6th edition, 5th centiles)

Parameter	Lower reference
Concentration	16 million/mL (total 39 million)
Progressive motility	30%
Total motility	42%
Volume	1.4 mL
Normal forms	4%
Vitality	54%

- These are reference values from fertile men, not pass/fail cut-offs. Severe oligozoospermia: under 5 million/mL.

Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- Hormonal: hypogonadotropism, exogenous testosterone or anabolic steroids, obesity (high oestradiol), hyperprolactinaemia.
- Testicular: cryptorchidism, mumps orchitis, torsion, chemotherapy or radiation; genetic (Klinefelter, AZF (azoospermia factor) c deletion).
- Motility: infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction, PCD (primary ciliary dyskinesia).
- Lifestyle: smoking, alcohol, cannabis, heat, gonadotoxins.

Evaluation

- **Two semen analyses** (2–7 days abstinence); history and examination (testis size, varicocele, vasa).
- Morning **testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone)**; oestradiol and prolactin if testosterone is low.
- Under 5 million/mL (especially with raised FSH or small testes): **karyotype and Y-microdeletion** (AUA/ASRM (American Urological Association/American Society for Reproductive Medicine) 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography) if volume low, DNA fragmentation.

Management

- **General**: stop smoking, weight loss, avoid heat; **stop exogenous testosterone**; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for palpable varicocele with abnormal semen; improves parameters and pregnancy. Not for subclinical ones.
- **Medical**: hypogonadotropic → hCG (human chorionic gonadotropin) with or without FSH. Idiopathic with low testosterone → off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if testosterone:oestradiol ratio under 10. Antioxidants: weak evidence.

- **ART (assisted reproductive technology)**: mild → IUI (intrauterine insemination); severe, or failed IUI → **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Spermatogenesis takes about 64 days, so judge response after 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicocele only; never give testosterone.

Q3b. Sperm retrieval techniques

Type: SAQ (10 marks)

Indications

- **OA (obstructive azoospermia)**: CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed or declined vasectomy reversal, or at the time of reconstruction (back-up).
- **NOA (non-obstructive azoospermia)**: testicular failure; only testicular retrieval works.
- Anejaculation (spinal cord injury): vibratory stimulation or electroejaculation first.
- All need **ICSI (intracytoplasmic sperm injection)**; cryopreserve spare sperm.

Techniques

Technique	Method	Best for	Drawback
PESA (percutaneous epididymal sperm aspiration)	Needle into caput under local	OA	Few sperm, may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened at 16–25×; micropipette by capillary action	OA (best)	Microsurgical skill
TESA (testicular sperm aspiration)	23 G needle, suction, many passes	OA	Immotile sperm, small yield; poor in NOA
TESE (testicular sperm extraction)	Open biopsy, one or several sites	OA, NOA	Blind sampling
Micro-TESE	Testis bivalved; larger opaque tubules picked under the microscope	**NOA (best)**	Long operation

- **MESA points**: move proximally until motile sperm appear (most motile in efferent ductules); 4–12 pipettes, frozen in aliquots.
- **Micro-TESE logic (Schlegel 1999)**: spermatogenesis in NOA is **focal**; tubules with sperm are larger and plumper than thin white Sertoli-cell-only tubules. Incision through an avascular plane, then the other testis if negative. It removes the least tissue for the most sperm.

Results

- OA: retrieval almost always succeeds.
- NOA micro-TESE: sperm in **about half**; better than conventional TESE.

Condition	Retrieval
Post-orchidopexy	74%
AZF (azoospermia factor) c deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
Sertoli-cell-only	30%
AZFa or AZFb deletion	**0% (do not operate)**

- NOA pregnancies have more miscarriage (sperm aneuploidy and DNA damage).

Complications

- Haematoma, infection, epididymal obstruction (PESA), testicular atrophy or transient fall in testosterone (TESE).

Also know: testicular sperm may beat ejaculated sperm when DNA fragmentation is high; postmortem retrieval only with prior consent; microfluidic sperm sorting.

Diagram to draw: micro-TESE with dilated tubules beside thin Sertoli-cell-only tubules.

> **Key points to remember:**
> - OA: epididymal (MESA); NOA: micro-TESE.
> - Micro-TESE finds sperm in about half of NOA.
> - AZFa/AZFb deletion: no sperm, do not operate.

Q3c. Treatment strategies for bilateral Wilm's tumour

Type: SAQ (10 marks)

Introduction

- **Synchronous bilateral WT (Wilms tumour)** is **5–7%**: **COG (Children's Oncology Group) stage V**. Each kidney is staged separately.
- Linked to WAGR (Wilms, aniridia, genitourinary anomalies, retardation), Denys–Drash and BWS (Beckwith–Wiedemann syndrome); **nephrogenic rests in over 90%**.
- Aim: **cure while preserving renal tissue**. Renal failure is about 15% at 15 years, mainly from bilateral nephrectomy.

Evaluation

- Ultrasound, contrast CT (computed tomography) or MRI (magnetic resonance imaging), chest CT; IVC (inferior vena cava) and coagulation checks.

COG approach (AREN0534)

```
VCR + AMD + DOX x 6 weeks (no upfront biopsy)
        |
   imaging at week 6
   /              \
NSS feasible     not feasible
 -> surgery       -> 6 more weeks chemotherapy
                    -> surgery by week 12
                    -> if poor response: open BILATERAL biopsy
```

- VCR = vincristine, AMD = actinomycin D (dactinomycin), DOX = doxorubicin; NSS = nephron-sparing surgery.
- **Why stop at 12 weeks**: in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders led to anaplasia. Non-shrinkage usually means differentiation, cured by excision.
- **Why open, bilateral biopsy**: anaplasia is discordant between kidneys in about 4 of 5.

Surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Enucleation: positive margins 23–36%, good outcome with radiotherapy.
- Unresectable both sides: bilateral nephrectomy, dialysis, transplant after **1–2 years** tumour-free.

Adjuvant

- Guided by worst histology and margins: anaplasia or positive margin → intensified chemotherapy and flank radiotherapy.

Outcomes

- AREN0534: surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, against 56% and 80.8% in NWTS-5.

SIOP (International Society of Paediatric Oncology)

- Same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

Follow-up: long term: blood pressure, proteinuria, creatinine, imaging (late relapse occurs).

Diagram to draw: bilateral tumours with planned partial nephrectomies.

> **Key points to remember:**
>
> - Stage V, 5–7%; no upfront nephrectomy or biopsy.
> - COG: three drugs for 6 weeks, surgery by week 12; SIOP: two drugs.
> - AREN0534: 4-year OS about 95%.

Q3d. Asymptomatic Hematuria

Type: SAQ (10 marks)

Definitions

- **MH (microhaematuria)**: **3 or more RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020). A dipstick alone is not enough (myoglobin, semen and peroxidases give false positives).
- **GH (gross haematuria)**: visible; exclude pigmenturia and menstrual blood.
- Prevalence about 6.5%. Malignancy: about 3% of MH; 14–26% of GH (DETECT I, IDENTIFY).

Causes

- **Neoplasm**: bladder (over 85% of cancers found), UTUC (upper tract urothelial carcinoma), RCC (renal cell carcinoma) about 1 in 1000 each, prostate.
- **Benign**: BPE (benign prostatic enlargement), stones, UTI (urinary tract infection), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal** (IgA nephropathy): dysmorphic RBCs, casts, proteinuria.
- **Others**: exercise, endometriosis, sickle cell. Anticoagulants do not explain haematuria; evaluate as usual.

Initial assessment

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch), examination, blood pressure, creatinine, culture.
- UTI → treat → repeat urinalysis; persistence means full evaluation. Glomerular picture → nephrology **and** urological evaluation.

AUA/SUFU risk stratification

	Low (all)	Intermediate (any)	High (any)
Age	Women under 50, men under 40	Women 50–59, men 40–59	60 or over
Smoking (pack-years)	Under 10	10–30	Over 30
RBCs/HPF	3–10	11–25	Over 25
Gross haematuria history	No	No	**Yes**
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT urography**

- CT (computed tomography) contraindicated → MR (magnetic resonance) urography, or retrograde pyelograms.
- Cytology and urine markers are **not** used initially; cytology if CIS (carcinoma in situ) is suspected.
- **Gross haematuria**: cystoscopy, CT urography and cytology at any age.

Follow-up

- Negative evaluation → repeat urinalysis within 12 months; release if negative.

- Re-evaluate for GH, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up.

> **Key points to remember:**
> - MH: 3 or more RBCs/HPF on microscopy, one sample.
> - Low: optional; intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.

Q3e. Prenatal and Neonatal assessment of neurogenic bladder dysfunction

Type: SAQ (10 marks)

Introduction

- Commonest cause: **myelomeningocele** (open NTD (neural tube defect)); also occult dysraphism, sacral agenesis, anorectal malformation.
- **Bony or motor level does not predict bladder function**, so every newborn needs testing.

Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound**: spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.
- **MOMS (Management of Myelomeningocele Study) 2011**: closure **before 26 weeks** reduced shunt need (about 40% vs 82%) and improved motor and cognitive scores, but caused prematurity (34 vs 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

Neonatal assessment (baseline)

Test	Timing / purpose
Examination	Spine, legs, anal tone, perineal sensation, bladder
Renal and bladder ultrasound	Before discharge, then 6-monthly for 2 years; hydronephrosis, wall thickness
Residual urine	Start CIC if not emptying
Creatinine	Within 3 months (cystatin C later)
VCUG (voiding cystourethrogram) or video-UDS	Reflux; dilating VUR (vesicoureteric reflux) in about 7%
UDS (urodynamics) with EMG (electromyography)	Within 3 months, once stable after closure
DMSA (dimercaptosuccinic acid)	Baseline scarring (EAU (European Association of Urology))

Urodynamic risk (UMPIRE (Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida))

Risk	DLPP (detrusor leak point pressure) or end-fill pressure
High (hostile)	40 cm H2O or more (McGuire 1981)
Intermediate	25–39, or detrusor overactivity
Low	Under 25

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.

Action

- High risk → **early CIC plus oxybutynin**: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) about 30% falls to under 2%.

- Repeat studies: the lesion can change (re-tethering).

Diagram to draw: neonatal UDS trace with DLPP over 40 cm H2O.

> **Key points to remember:**
> - Level does not predict bladder; UDS within 3 months.
> - DLPP 40 cm H2O or more = hostile; early CIC + antimuscarinic.
> - MOMS: fewer shunts, no bladder benefit.

Q3f. Marginal Donor

Type: SAQ (10 marks)

Definition

- A donor whose kidney carries **more DGF (delayed graft function) and shorter graft survival** than a standard donor; used because of organ shortage.

Categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **60 or over**, or **50–59 with 2 or more** of: hypertension, **creatinine over 1.5 mg/dL**, death from **CVA (cerebrovascular accident)**. Graft-loss risk about 70% higher.

2. KDPI (Kidney Donor Profile Index) over 85%: replaced ECD in the US (2014). **Longevity matching**: lowest-KDPI kidneys go to recipients with the best EPTS (Estimated Post-Transplant Survival).

3. DCD (donation after circulatory death), Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of support	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled

- Warm ischaemia raises DGF (roughly 40–50%); long-term survival is similar to DBD (donation after brain death).

4. Others: AKI (acute kidney injury) donors; **HCV (hepatitis C virus)-viraemic** donors with antivirals; **paediatric donors under 15 kg (en bloc)**; older living donors.

- **Excluded**: metastatic or haematological cancer, melanoma, untreatable viral infection.

Assessment and use

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, vessels, 0–3 each): **0–3 single**, **4–6 dual kidney transplant**, **7 or more discard**.
- **HMP (hypothermic machine perfusion)** reduces DGF versus cold storage (Moers, 2009), without long-term survival difference; normothermic perfusion tests viability.
- **ESP (Eurotransplant Senior Program)**: donors 65 or over to recipients 65 or over, with short cold ischaemia.
- Recipient: informed consent; induction and delayed CNI (calcineurin inhibitor) if DGF.

Urological relevance

- Older and ECD kidneys have more **urine leak and ureteric stricture** (ischaemic distal ureter), as do multiple arteries and DGF.
- **Ojo (2001)**: ECD transplantation improves survival over staying on dialysis.

Diagram to draw: Maastricht categories on a timeline of circulatory arrest.

> **Key points to remember:**
> - ECD: 60 or over, or 50–59 with 2 of hypertension, creatinine over 1.5, CVA death.
> - Remuzzi 0–3 single, 4–6 dual, 7 or more discard.
> - HMP reduces DGF; Maastricht III is the commonest DCD.

Rapid Revision Sheet – 2016-Summer Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss minimally invasive and endoscopic management of upper urinary tract malignancy. (LAQ)

- EAU low risk needs all: unifocal, under 2 cm, low grade, no invasion; any single adverse feature makes it high risk.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).

Q2. Laparoscopic pyeloplasty. (LAQ)

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

Q3a. Oligoanaesthospermia. (SAQ)

- WHO 2021: concentration 16 million/mL, progressive motility 30%.
- Karyotype and Y-microdeletion if under 5 million/mL.
- Repair palpable varicocele only; never give testosterone.

Q3b. Sperm retrieval techniques (SAQ)

- OA: epididymal (MESA); NOA: micro-TESE.
- Micro-TESE finds sperm in about half of NOA.
- AZFa/AZFb deletion: no sperm, do not operate.

Q3c. Treatment strategies for bilateral Wilm's tumour (SAQ)

- Stage V, 5–7%; no upfront nephrectomy or biopsy.
- COG: three drugs for 6 weeks, surgery by week 12; SIOP: two drugs.
- AREN0534: 4-year OS about 95%.

Q3d. Asymptomatic Hematuria (SAQ)

- MH: 3 or more RBCs/HPF on microscopy, one sample.
- Low: optional; intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.

Q3e. Prenatal and Neonatal assessment of neurogenic bladder dysfunction (SAQ)

- Level does not predict bladder; UDS within 3 months.
- DLPP 40 cm H2O or more = hostile; early CIC + antimuscarinic.
- MOMS: fewer shunts, no bladder benefit.

Q3f. Marginal Donor (SAQ)

- ECD: 60 or over, or 50–59 with 2 of hypertension, creatinine over 1.5, CVA death.
- Remuzzi 0–3 single, 4–6 dual, 7 or more discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.